

MOHAMED BADAWY

Java Backend Developer | Flutter Developer | Minecraft Plugin Developer | IoT & Embedded Systems Engineer
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PROFESSIONAL SUMMARY

Java backend developer and Flutter engineer with hands-on experience architecting high-concurrency, scalable server systems (Spigot, Paper, Velocity, BungeeCord) and cross-platform mobile applications. Skilled in distributed systems, REST API integration, SQL/NoSQL data modeling, and performance optimization. Actively growing through competitive programming and IEEE ECU leadership roles.

EXPERIENCE

Java Plugin Developer — DevRoom · Italy · Remote

March 2026 – Present

- Architected and maintained high-performance Java server systems (Spigot/Paper/Velocity), covering backend logic, database design (MongoDB, MySQL), cross-server networking, and REST API/webhook integrations.
- Client feedback:** 8 commissions completed with a 5.0/5.0 average rating.

Java Plugin Developer — Rollerite LLC · United States · Remote

October 2025 – Present

- Built and maintained modular Java plugin systems (Spigot/Paper/BungeeCord) for clients, handling the full lifecycle from requirements to deployment, with SQL/NoSQL data modeling and distributed system design.
- Client feedback:** 4.83/5.00 average rating (6 reviews).

FEATURED PROJECTS

Projecto-Messio — Team Project

- Smart home automation system combining ESP32 hardware control, a C++ backend, and a Flutter mobile app; built full-stack integration handling real-time hardware-to-app communication for smart home devices. **Skills:** ESP32/embedded programming, C++, Flutter, real-time communication protocols, team collaboration.

HealthLink App Suite — Team Project

- Multi-platform healthcare application spanning a Flutter mobile app, a Flutter desktop app, and a C++ backend, delivering a consistent experience across mobile and desktop with shared backend logic. **Skills:** Flutter (mobile & desktop), C++ backend development, cross-platform architecture, team collaboration.

PunishmentSystem

- High-performance Minecraft punishment management plugin (bans, mutes, kicks, IP-bans) with async database operations and an intelligent selective-caching system, reducing memory usage by 80% and achieving sub-1ms mute checks with 95%+ cache hit rate. Supports SQLite, MySQL, and MongoDB, plus a custom API and event system for third-party plugin integration. **Skills:** Java, Spigot/Paper API, async programming, caching strategy design, multi-database architecture (SQL & NoSQL), API/event system design.

Agricultural Monitoring and Automation Enabler — IEEE ITC-Egypt 2025

- Feedback-based desktop remotely controlled greenhouse monitoring system, published as a peer-reviewed IEEE paper. **Skills:** Embedded systems programming, sensor integration, real-time control systems, academic research.

RESEARCH

Published Research Paper — IEEE ITC-Egypt 2025

"Agricultural Monitoring and Automation Enabler: Feedback-Based Desktop Remotely Controlled System"

Co-authored with Noha Hussen, Omar Moustafa, and Salah Othman | Published in: 2025 International Telecommunications Conference (ITC-Egypt), IEEE, Cairo, Egypt | DOI: 10.1109/ITC-Egypt66095.2025.11186572

- Designed an energy-efficient, automated feedback-based greenhouse monitoring system to address climate-driven agricultural challenges in Egypt, integrating an Arduino microcontroller with DHT11 (temperature/humidity) and LDR (light intensity) sensors for real-time environmental control.
- Demonstrated strong academic research capability from methodology design through publication. **Skills:** Embedded systems programming (Arduino), sensor integration, academic research & technical writing, IEEE-standard documentation, environmental data analysis.

VOLUNTEERING

Vice Head of Public Relations — IEEE ECU Student Branch

September 2025 – Present

- Directing public relations strategies and managing outreach initiatives to enhance the branch's overall visibility and engagement.

Data Structures Instructor — IEEE ECU Student Branch

December 2025 – January 2026

- Instructed peer students on core C++ data structures and fundamental concepts through comprehensive academic review sessions.

Public Relations Specialist — IEEE ECU Student Branch

July 2025 – September 2025

- Facilitated internal and external communications while supporting key PR campaigns and organizational activities.

Event Organizer — IEEE ECU Student Branch

July 2025 – Present

- Coordinated large-scale events hosted at the Egyptian Chinese University (ECU), ensuring smooth execution and logistical support for:
 - **IEEE Egyptian Student Paper Contest (ESPC) 2025:** A national research contest gathering student branches from across Egypt.
 - **Made in Egypt (MIE) Closing Ceremony 2025:** An IEEE Egypt Section initiative bridging academia and industry through graduation projects.

EDUCATION

Bachelor of Engineering, Software Engineering & Information Technology

Faculty of Engineering and Technology, Egyptian Chinese University | 2024 – 2029 (In Progress) | GPA: 3.74/4.0

CERTIFICATIONS

Certificate of Participation and Presentation (ITC Conference Paper) — ITC-EGYPT ADC 2026

ITC-EGYPT ADC 2026

EF SET English Certificate — EF Standard English Test

68/100 (C1 Advanced)

Problem Solving Certificate — Nile University

Competitive Programming Arena (NUCPA) 2026

Mobile Dev Flutter — IEEE FCIH SB

IEEE Student Branch

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TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Flutter, Dart, C++

Server/Game APIs: Spigot, Paper, Velocity, BungeeCord

Databases: MongoDB, MySQL, SQLite, PostgreSQL, Firebase Firestore

Architecture & Practices: REST APIs, Webhooks, Clean Design Patterns, Multi-threading, MVVM, MVC

Embedded/IoT: Arduino, ESP32, Sensor Integration

Research: Academic Research & Technical Writing, IEEE Publication Standards

Tools: Git, Gradle, Maven, Android Studio, IntelliJ IDEA, VS Code, Figma

Competitive Programming: Codeforces, LeetCode — Data Structures & Algorithms

LANGUAGES

Arabic — Native | English — C1 Advanced (EF SET Certified)